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Customer-Base Concentration and Inventory Efficiencies: Evidence from the Manufacturing Sector

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What is the link between customer-base concentration and inventory efficiencies in the manufacturing sector? Using hand-collected data from 10-K Filings, we find that manufacturers with more concentrated customer bases hold fewer inventories for less time and are less likely to end up with excess inventories, as indicated by the lower likelihood and magnitude of inventory write-downs and reversals. Using disaggregated inventory disclosures, we find that inventory efficiencies primarily flow through the finished goods inventory account, while raw material efficiencies are offset by higher work-in-process holdings and longer work-in-process cycles. In additional analysis, we document a valuation premium for more concentrated manufacturers after controlling for other firm characteristics, including default risk and cost of capital estimates. We conclude that investors trade off the costs and benefits of relationships with a limited number of major customers and, on balance, consider customer-base concentration as a net positive for firm valuation. Overall, our study adds to interdisciplinary research in accounting and operations management by shedding new light on the relevance of major customer disclosures for fundamental analysis and valuation in the manufacturing sector.

Key words: Customer-base concentration; inventory efficiencies; inventory write-downs; firm valuation
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1. Introduction

Although inventory management is as old as manufacturing itself, little is known about the empirical link between customer-base structure and inventory efficiencies. This link is *ex ante* unclear. On one hand, the conventional view based on the relative bargaining power literature emphasizes that manufacturers with concentrated customer bases are held captive by their major customers and are forced to carry excess inventory (*bargaining power view*). On the other hand, researchers in operations management (OM) argue that relationships with a limited number of major customers enable manufacturers to achieve inventory efficiencies through the implementation of supply-chain practices that mitigate demand uncertainty (*OM view*).

In this study, we investigate the link between customer-base concentration (*CC*) and inventory efficiencies for a panel of manufacturers in the United States. We focus on manufacturers because inventories account for a significant fraction of such firms' asset base. Given that inventory is costly to hold and is subject to obsolescence risk, one of the main objectives of inventory management is mitigating demand uncertainty, which translates into lower inventory holdings, shorter inventory holding periods, and lower obsolescence risk. According to industry experts, firms with lower inventories are more valuable than firms with bloated inventories (e.g., Chen et al. 2005, Lieberman et al. 1999). Indeed, studies in fundamental analysis and valuation provide evidence that firms with large inventory changes (i.e., inventory accruals) underperform in the stock market (e.g., Allen et al. 2013, Belo and Lin 2012, Thomas and Zhang 2002).

Consistent with the OM view, we find that manufacturers with more concentrated customer bases hold significantly fewer inventories and experience shorter inventory holding periods. Our findings hold after controlling for a wide array of firm characteristics, including firm size, age, growth, sales volatility, number of reported business segments, leverage, margins, and capital intensity, as well as industry and year fixed effects. Following the accounting cost flows, we disaggregate total inventories into raw materials, work-in-process, and finished goods. Our granular investigation of individual inventory accounts draws a nuanced picture of the link between *CC* and inventory efficiencies. Specifically, we find that inventory efficiencies primarily flow through the finished goods inventory account, while raw material efficiencies are offset by higher work-in-process holdings and longer work-in-process cycles.

Data Availability: Data are available from the sources identified in the text.

To examine the link between *CC* and inventory investment efficiencies, we hand-collect information about the occurrence and magnitude of inventory write-downs. Under U.S. GAAP firms are required to write down inventories, that is, reduce the balance sheet carrying value, when market value falls below reported cost. Inventory write-downs are dictated by the lower of cost or market basis for inventory valuation, while inventory write-ups are generally not allowed. Inventory write-downs represent a common way in which prior accrual estimation errors are reversed (e.g., Allen et al. 2013), and can offer insights into a firm's propensity to end up with excess inventory without reference to a model for the optimal level of inventory investment.

Although inventory write-down information is not readily available from the usual academic data sources, we are able to extract such information directly from 10-K filings using a text mining algorithm. Our algorithm searches the 10-K filings available post-1995 through the SEC EDGAR database for variations in keyword strings and extracts all matches. We manually review all matches and read the corresponding 10-K filings to confirm whether or not an inventory write-down was recorded and, if so, its dollar amount. Our hand-collected sample of reported inventory write-downs yields evidence consistent with a positive link between *CC* and inventory investment efficiencies.

Using logistic regressions, we find that more concentrated manufacturers are significantly less likely to report an inventory write-down. Turning to the magnitude of reported write-downs, we use Tobit regressions (due to left censoring of write-downs at zero) and find that conditioning on the occurrence of an inventory write-down its absolute magnitude decreases with *CC*. Overall, our evidence suggests that more concentrated manufacturers are associated with a lower likelihood and magnitude of inventory write-downs, which is consistent with the OM view that such firms are less likely to end up with excess inventories.

To gain insights for our full sample period (including the years prior to the introduction of the SEC EDGAR database), we search for a link between *CC* and inventory reversals, which we define as a decline in total inventories in excess of 1% of the beginning of year total assets accompanied by commensurate negative Compustat special items. After we confirm that inventory reversals and inventory write-downs share a common component, we provide evidence that relationships with a limited number of major customers mitigate accrual estimation errors: more concentrated manufacturers are associated with a lower likelihood and magnitude of inventory reversals.

After establishing a significantly positive link between *CC* and inventory efficiencies, we test whether there is a valuation premium for more concentrated manufacturers. If *CC* translates into value-relevant efficiencies, then Tobin's *Q*, that is, the ratio of the market value of invested capital scaled by the book value of total assets, should be higher for more concentrated manufacturers. Consistent with this prediction, we document a valuation premium for more concentrated manufacturers after controlling for other firm characteristics, including default risk and cost of capital estimates. We acknowledge that our evidence of a valuation premium may only be partially attributed to that between *CC* and inventory efficiencies. Nevertheless, the evidence suggests that investors trade off the costs and benefits of relationships with a limited number of major customers and, on balance, consider *CC* as a net positive for firm valuation.

We contribute to accounting and OM research in several ways. First, we focus on the manufacturing sector and provide direct evidence on the link between *CC* and inventory efficiencies. Second, using disaggregated inventory disclosures, we shed light on how inventory efficiencies accrue to more concentrated manufacturers. Third, using our hand-collected sample of inventory write-downs as well as Compustat information about reported special items, we uncover a link between *CC* and inventory investment efficiencies. Fourth, our evidence of a valuation premium for more concentrated manufacturers implies that investors consider relationships with a limited number of major customers as a net positive for firm valuation. More generally, our study provides new insights on the costs and benefits of major customer relationships for supplier firms—a topic that is of growing significance among academics and practitioners alike (e.g., Banerjee et al. 2008, Cachon and Olivares 2010, Campello and Gao 2014, Cen et al. 2015, Dhaliwal et al. 2015, Irvine et al. 2014, Kale and Shahrur 2007, Krishnan et al. 2015, Osadchiy et al. 2015, Patatoukas 2012).

The study proceeds as follows. Section 2 provides the background and motivates our research questions. Section 3 describes the research design. Section 4 presents the results. Section 5 concludes and provides directions for future research.

2. Background and Research Questions

What is the link between *CC* and inventory efficiencies? In what follows, we discuss the *bargaining power view*, which predicts a negative association between *CC* and inventory efficiencies, and the *OM view*, which predicts a positive association between *CC* and inventory efficiencies. We conclude that the direction

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of the link between inventory efficiencies and *CC* is *ex ante* unclear.

2.1. The Bargaining Power View

The bargaining power view emphasizes the negatives of dealing with a limited number of major customers (e.g., Galbraith 1952, Lustgarten 1975, Porter 1974, Scherer 1970). The conventional argument goes as follows. Major customers have the bargaining power and pressure their dependent suppliers to maintain a high level of product availability, which is typically measured in terms of fill rates, that is, the fraction of customer demands that can be filled from inventory immediately, and in-stock probability, that is, the probability that all customer demands are satisfied over a time interval. In turn, higher fill rates and in-stock probability translate into higher inventory holdings and longer inventory holding periods (e.g., Cachon and Terwiesch 2008). According to the bargaining power view, major customers may also use their power to pressure their dependent suppliers to agree to carry on their books finished goods that would normally belong to the customers. An example of such agreement is consignment inventory, under which the manufacturer retains ownership of products located at customer sites until the customer consumes or resells the products (e.g., Kelly and Gosman 2000).

Overall, the bargaining power view predicts a negative association between *CC* and inventory efficiencies. Based on this view, more concentrated manufacturers are predicted to hold more inventories and to experience longer inventory holding periods. Relatedly, more concentrated manufacturers are predicted to be more likely to end up with excess inventory on their books with increased obsolescence risk, which should translate into more frequent and larger inventory write-downs.

2.2. The OM View

Although the bargaining power view emphasizes the negatives of *CC*, research in OM suggests that relationships with a limited number of major customers enable the implementation of supply-chain practices that mitigate demand uncertainty thereby leading to inventory efficiencies. One such practice is collaborative planning, forecasting, and replenishment (CPFR), which is a set of technology-enabled standards that provides a roadmap for enabling collaborative demand and supply planning and execution processes (e.g., Aviv 2001). A key component of CPFR is information sharing, under which the manufacturer sets up a common information system with its major customers to exchange demand forecast updates (e.g., Ren et al. 2010, Terwiesch et al. 2005). Such infrastructure is set up to reduce demand uncertainty and mitigate supply-chain distortions such as the bullwhip effect, that is, the phenomenon where orders to the upstream supplier tend to have higher variance than sales to the downstream customer (e.g., Lee et al. 1997). In turn, enhanced demand forecasting leads to a reduction of inventory investment because the optimal buffer size of the safety stock is an increasing function of demand uncertainty (e.g., Porteus 2002, Zipkin 2000).

Another supply-chain practice is known as vendor-managed inventory (VMI). Under a VMI agreement, the manufacturer makes the main inventory replenishment decisions for its customer. This means that the customer relinquishes control of key resupply decisions, while the manufacturer monitors the customer's inventory levels and makes periodic replenishment decisions regarding order quantities, shipping, and timing (e.g., Kulp 2002). Suppliers are attracted to VMI because coordination and information sharing mitigates demand uncertainty and helps dampen the peaks and valleys of production, thereby allowing smaller buffers of capacity and inventory. Trust between the supply-chain partners is critical for the successful implementation of VMI agreements. Therefore, such agreements are more likely to be implemented successfully when the supplier is serving only a limited number of major customers (e.g., Kulp et al. 2004, Vollmann and Cordon 1998, Waller et al. 1999).

Relationships with a limited number of major customers may also enable the successful adoption of just-in-time (JIT) manufacturing. JIT is characterized as a pull approach to manufacturing, as opposed to the traditional push approach (e.g., Selto et al. 1995). Under JIT, manufacturers produce in smaller batches only in response to a customer order so that inventories are minimized. This is in contrast to the traditional push approach to manufacturing where inventories build up if parts are not used and products are not sold as quickly as they can be produced. Indeed, prior studies provide evidence that JIT manufacturing reduces inventory levels and increases inventory turns (e.g., Fullerton et al. 2003, Huson and Nanda 1995). Balakrishnan et al. (1996) propose that major customers often urge for the adoption of JIT manufacturing in a concerted effort to help their dependent suppliers reduce inventory levels. Relatedly, Kinney and Wempe (2002) argue that although JIT manufacturing requires substantial downstream coordination costs, such costs are decreasing in the number of major customers. Therefore, more concentrated manufacturers enjoy lower coordination costs in ordering, production, scheduling, and delivery.

Overall, the OM view predicts a positive link between *CC* and inventory efficiencies. In particular,

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more concentrated manufacturers are predicted to hold fewer inventories and to experience shorter inventory holding periods. The OM view also predicts that more concentrated manufacturers are less likely to end up with excess inventories on their books, therefore the likelihood and magnitude of inventory write-downs are predicted to decrease with *CC*.

2.3. Null Hypothesis

On balance, we conclude that the link between *CC* and inventory efficiencies is *ex ante* unclear. On one hand, increased major customer dependency may be a reflection of an adversarial supply-chain relationship in which the manufacturer is pressured to carry extra inventories with increased obsolescence risk. On the other hand, relationships with a limited number of major customers may enable manufacturers to achieve inventory efficiencies through the implementation of supply-chain practices that mitigate demand uncertainty. Therefore, we state our hypothesis in null form: *there is no association between CC and inventory efficiencies*.

3. Research Design

The starting point in our search for a link between customer-base structure and inventory efficiencies in the manufacturing sector is Patatoukas' (2012) measure of *CC*:

$$CC_{it} = \sum_{j=1}^{J} \left(\frac{Sales_{ijt}}{Sales_{it}} \right)^2, \tag{1}$$

where $Sales_{ijt}$ represents firm i's sales to major customer j in year t and $Sales_{it}$ represents firm i's total sales in year t. The *CC* measure captures two elements of diversification, namely the number of major customers with which the firm interacts, and the relative importance of each major customer in the firm's annual revenue. The range of *CC* is between 0 and 1, where lower (higher) values correspond to less (more) concentrated customer bases. Major customer disclosures are mandated by Statement of Financial Accounting Standards No. 131 of the Financial Accounting Standards Board and Regulation S-K of the Securities and Exchange Commission.

3.1. Customer-Base Concentration and Inventory Efficiencies

To examine the link between *CC* and inventory efficiencies, we start with inventory variables commonly used in financial statement analysis, including inventory holding ratios and average inventory days outstanding. We measure inventory holdings as the ratio of total inventories scaled by total assets (*TOTHLD*). We measure average inventory days outstanding as the ratio of total inventories scaled by cost of goods sold multiplied by 365 (*TOTDAYS*). We search for a link between inventory efficiencies and *CC* using OLS regression models of the following form:

$$\text{Inventory Efficiencies}_{it} = a + \beta_1 CC_{it} + \sum_{k=2}^{K} \beta_k X_{it}^k + \varepsilon_{it}. \tag{2}$$

Throughout the study, we estimate models using pooled cross-sectional regressions. As explained in Section 2, our null hypothesis is no association between *CC* and inventory efficiencies. Therefore, we use two-tailed tests when testing for statistical significance. It is well known that OLS standard errors are biased when the residuals of a given firm are correlated across years or the residuals of a given year are correlated across firms. To address time-series and cross-sectional residual dependence, we base statistical inferences on two-way cluster-robust standard errors (e.g., Petersen 2009).[1]

The vector of control variables X^k includes log market capitalization (*MV*), log firm age (*AGE*), sales growth (*SG*), sales growth volatility (*SGV*), and an indicator variable for manufacturers with at least two business segments (*CONGLO*). These variables are included because more concentrated firms tend to be smaller, with higher sales growth, and are less likely to operate in multiple business segments (Patatoukas 2012). We include total leverage (*LEV*) to control for variation in capital structure. We include gross margins (*GM*) to control for the "earns-vs.-turns" tradeoff, that is, the managerial practice of trading off inventory turns with gross margins (e.g., Gaur et al. 2005). To control for the impact of accumulated losses, we include an indicator variable for manufacturers with either negative book value of equity or book value of equity in excess of market value of equity (*ACCLOSS*). We include the ratio of net PP&E to total assets (*CAP*) to control for variation in capital intensity, which is known to be positively related with inventory turnover (e.g., Gaur and Kesavan 2009). The vector of control variables also includes industry and year fixed effects. Industry fixed effects are based on two-digit SIC codes and are included because industry membership is a source of variation in firm performance and inventory practices (e.g., Blazenko and Vandezande 2003, Radhakrishnan et al. 2014). Year fixed effects are based on the calendar year of fiscal year-ends and are included to control for aggregate time-varying factors affecting inventory efficiencies (e.g., Rumyantsev and Netessine 2007).

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3.1.1. Disaggregated Inventory Accounts. Manufacturers record three accounts of inventories: raw materials, work-in-progress, and finished goods. The raw materials account absorbs the cost of parts and materials purchased from vendors for use in the production process. When raw materials are used in the production process, the cost of the materials used is transferred into work-in-process inventory. The work-in-process account absorbs the cost of the inventory of partially completed goods. Work-in-process includes the materials used in the production as well as labor cost and overhead cost. When the production process is completed, the cost of goods is transferred from work-in-process to the finished goods account. In turn, the finished goods account absorbs the cost of the stock of completed goods ready for delivery to customers. When finished goods are sold, cost of goods sold is debited and finished goods inventory is credited. Following the accounting cost flows, we disaggregate total inventories and investigate the link between *CC* and inventory efficiencies across all three inventory accounts. The total inventories are reported on the balance sheet, while the disaggregated inventory account balances are typically disclosed in the notes to the financial statements. Given that the supplier–customer relationship is centered on the transfer of ownership of finished goods, the OM view predicts that inventory efficiencies are likely to flow mostly through this inventory account.

3.1.2. LIFO Reserves and Financial Ratio Comparability. A common chore in financial statement analysis is converting the results of a firm that uses "last-in, first-out" (LIFO) for inventory costing so that they can be compared with a firm that uses "first-in, first-out" (FIFO). Achieving comparability in inventory ratios across firms is possible because firms that use LIFO are required to disclose what their total inventories would have been had they used FIFO. The excess of FIFO value over LIFO is known as the LIFO reserve. We use LIFO reserve information from Compustat and make the following adjustments to the reported numbers:

$$\begin{aligned} &Total\ Inventories_{it}^{FIFO} = Total\ Inventories_{it}^{LIFO} \\ &+ LIFO\ RESERVE_{it}, Total\ Assets_{it}^{FIFO} = Total\ Assets_{it}^{LIFO} \\ &+ LIFO\ RESERVE_{it}, COGS_{it}^{FIFO} = COGS_{it}^{LIFO} \\ &- (LIFO\ RESERVE_{it} - LIFO\ RESERVE_{it-1}). \end{aligned} \tag{3}$$

We note that LIFO reserve disclosures are available only for total inventories. To adjust individual inventory accounts, we allocate the LIFO reserve based on the relative contribution of raw materials, work-in-process, and finished goods to total inventories. Our results are not sensitive when we allocate the LIFO reserve entirely to raw materials or finished goods.

3.2. Customer-Base Concentration and Inventory Write-Downs

U.S. GAAP requires firms to value inventories at the lower of cost or market (*Accounting Standards Codification 330 Inventory*). Market value is defined as current replacement cost, but not greater than net realizable value (estimated as selling price less reasonable costs of completion and disposal) and not less than net realizable value reduced by a normal profit margin. Under the lower of cost or market basis for inventory valuation, inventory write-downs are required when market value falls below reported cost, while inventory write-ups are generally not allowed when market value rises above reported cost. Under U.S. GAAP, any write-down of inventory to the lower of cost or market creates a new cost basis that subsequently cannot be reversed. Inventory write-downs under U.S. GAAP represent a common way in which prior inventory accrual estimation errors are reversed (e.g., Allen et al. 2013) and can offer insights into a firm's propensity to end up with excess inventory without reference to a model for the optimal level of inventory investment.[2]

According to the Compustat North America User's Guide, inventory write-downs are reclassified by Compustat in special items (mnemonic SPI) when reported separately or called nonrecurring. Compustat special items, however, are a "black box" pooling inventory write-downs with as many as 23 categories of unusual or nonrecurring items (e.g., Frankel 2009). To circumvent the limitations of Compustat special items, we develop a text mining algorithm to extract inventory write-down information directly from 10-K filings, which are widely available post-1995 on the SEC EDGAR database (e.g., Griffin 2003).

Our text-mining algorithm searches the 10-K filings for the strings "writ" and "wrot," which capture variations in the word write, within 15 words of the string "inventor," which captures variation in the word inventory. We manually review the matches and read all corresponding 10-K filings to confirm whether or not an inventory write-down was recorded and, if so, its dollar amount. As an additional sanity check, we identify observations associated with a decline in total inventories in excess of 1% of the beginning of year total assets and if the text mining algorithm yields no match, then we manually review the corresponding 10-K filings.

We examine the association between *CC* and the occurrence of inventory write-downs using the following logistic regression model:

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$$DWD_{it} = a + \beta_1 CC_{it} + \sum_{k=2}^{K} \beta_k X_{it}^k + \varepsilon_{it}, \qquad (4)$$

where DWD_{it} is an indicator variable that equals one if a firm reports an inventory write-down. The coefficient of interest is the slope on *CC*. A negative (positive) β_1 coefficient would suggest that more concentrated manufacturers are less (more) likely to end up with excess inventory, as indicated by the lower (higher) likelihood of an inventory write-down. The vector of control variables is the same as the one defined in equation (2).

We also search for an association between *CC* and the absolute magnitude of inventory write-downs (*WD*). Due to the left censoring of *WD* at zero, we estimate the following Tobit regression model:

$$WD_{it} = a + \beta_1 CC_{it} + \sum_{k=2}^{K} \beta_k X_{it}^k + \varepsilon_{it}. \qquad (5)$$

To shed light on the association between inventory investment efficiencies and *CC* for our full sample period including the period prior to the introduction of the SEC EDGAR database, we use Compustat special items to construct a proxy for inventory write-downs and repeat the regressions in equations (4) and (5). Specifically, we construct an indicator variable, denoted *DREVERSAL*, that equals one if a firm experienced an inventory reversal. We define an inventory reversal as a decline in total inventories in excess of 1% of the beginning of period total assets accompanied by commensurate negative Compustat special items. The key advantage of *DREVERSAL* is that it can be measured for our full sample period, while *DWD* can be measured only post-1995, that is, after the introduction of the SEC EDGAR database. Focusing on the post-1995 period, we find that inventory reversals and inventory write-downs share a common component with a correlation coefficient of 71%.

3.3. Customer-Base Concentration and Firm Valuation

If relationships with a limited number of major customers translate into value-relevant efficiencies, then more concentrated manufacturers should trade at a higher valuation premium. To shed light on the valuation link, we search for an association between the Tobin's *Q* ratio, that is, the ratio of the market value of invested capital scaled by the book value of total assets, and *CC*. Specifically, we estimate the following OLS regression model:

$$Tobin's\ Q_{it} = a + CC_{it} + \sum_{k=2}^{K} \beta_k X_{it}^k + \varepsilon_{it}. \qquad (6)$$

According to standard valuation models (e.g., Penman 2011), the Tobin's *Q* is (i) increasing in expected growth and profitability and (ii) decreasing in discount rates. Specifically, a higher Tobin's *Q* implies that investors assign a higher valuation premium because they are expecting higher growth, and/or higher profitability, and/or lower discount rates. If having a limited number of major customers is a net positive (negative) for the valuation of manufacturers, then the slope on *CC* should be positive (negative).[3]

3.4. Sample

Our study focuses on U.S. manufacturers with SIC codes between 2000 and 3999. For a firm-year to be included in our sample it must have nonmissing data about *CC*, inventory holding components, inventory holding periods, inventory reversals, and the vector of control variables. To reduce the impact of extreme observations, we trim variables at the top and bottom one percentile based on annual cross-sectional cutoffs. Our sample includes 15,084 firm-year observations from 1977 to 2006, spanning 20 industries in the manufacturing sector. The sample covers the 30-year period studied by Patatoukas (2012) and starts in 1977 because this is the first year of Compustat coverage of major customer disclosures. Post-1995, the sample includes 7893 firm-year observations.

4. Empirical Results

4.1. Descriptive Analysis

Table 1, Panel A, reports descriptive statistics. We find evidence of significant variation in the level of *CC*. The average value of *CC* is 0.113 with a standard deviation of 0.168. Figure 1a shows that *CC* rises from an average value of 0.06 in 1977 to 0.13 in 2006. Evidence of increasing *CC* overlaps with increasing inventory efficiencies over the sample period. Prior to turning to inventory efficiency variables, we note that the majority of manufacturers do not report LIFO reserves. Indeed, the mean value of the indicator variable *LIFO* shows that such reserves are reported for 14.5% of our sample. Figure 1b provides evidence that LIFO usage commoves over time with the GDP price deflator, which is consistent with the fact that firms have more incentives to adopt the LIFO method for inventories when inflation is high (e.g., Biddle and Ricks 1988). The frequency of reported LIFO reserves peaked in 1981 at 30% and has dropped to 9% by 2006. We observe that inflation also peaked in 1981 at 9.3% and has hovered around 2.5% since the mid-1980s.

The descriptive statistics for inventory holdings highlight the importance of inventory management in the manufacturing sector. The average manufacturer ties up as much as 21.5% of its asset base in inventories.

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Table 1 Descriptive Statistics

			Percentiles				
	Mean	SD	5th	25th	50th	75th	95th
Panel A: Empirical distributions							
CC_{it}	0.113	0.168	0.003	0.018	0.047	0.130	0.470
$LIFO_{it}$	0.145	0.352	0.000	0.000	0.000	0.000	1.000
$TOTHLD_{it}$	0.215	0.141	0.008	0.107	0.196	0.307	0.469
$FINHLD_{it}$	0.077	0.086	0.000	0.017	0.049	0.104	0.261
$WIPHLD_{it}$	0.055	0.066	0.000	0.009	0.034	0.077	0.188
$RAWHLD_{it}$	0.083	0.072	0.000	0.029	0.066	0.117	0.226
$TOTDAYS_{it}$	113.4	80.8	8.1	61.6	98.1	148.2	260.7
$FINDAYS_{it}$	38.6	40.5	0.0	10.1	27.4	54.9	114.1
$WIPDAYS_{it}$	31.1	41.7	0.0	5.0	17.7	41.0	106.8
$RAWDAYS_{it}$	43.7	41.8	0.0	16.6	33.1	59.5	119.2
$DREVERSAL_{it}$	0.089	0.285	0.000	0.000	0.000	0.000	1.000
DWD_{it}	0.058	0.234	0.000	0.000	0.000	0.000	1.000
$TOBIN_{it}$	1.800	1.999	0.491	0.793	1.186	2.020	5.090

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Panel B: Pairwise correlations						
(1) CC_{it}	1.000	−0.190***	−0.137***	−0.027***	−0.045***	0.187***
(2) $TOTHLD_{it}$		1.000	0.471***	−0.001	0.049***	−0.240***
(3) $TOTDAYS_{it}$			1.000	−0.040***	0.021*	0.005
(4) $DREVERSAL_{it}$				1.000	0.713***	−0.062***
(5) DWD_{it}					1.000	−0.082***
(6) $TOBIN_{it}$						1.000

		CC_{it} quintile portfolio		
		Q_1 (Low)	Q_5 (High)	Spread $Q_5 - Q_1$
Panel C: Univariate comparisons				
$TOTHLD_{it}$	Mean	0.216	0.180	−0.036***
	Median	0.195	0.154	−0.042***
$TOTDAYS_{it}$	Mean	118.9	95.8	−23.1***
	Median	103.0	79.9	−23.1***
$DREVERSAL_{it}$	Mean	0.094	0.072	−0.022***
	Median	0.000	0.000	0.000
DWD_{it}	Mean	0.066	0.032	−0.033***
	Median	0.000	0.000	0.000
$TOBIN_{it}$	Mean	1.640	2.312	0.672***
	Median	1.123	1.469	0.346***

This Table reports descriptive statistics for key variables. The variables are defined in Appendix A. Panel A reports the empirical distributions. Panel B reports pairwise correlations. Panel C reports the mean and median values of key variables for the top (Q_5) and bottom (Q_1) quintile portfolio based on customer-base concentration CC_{it} along with the spread in the mean and median values. ***, **, and * indicate statistical significance at 1, 5, and 10 levels, respectively, using two-tailed tests. The sample includes 15,084 firm-year observations from 1977 to 2006. Post-1995, the sample includes 7893 firm-year observations.

The disaggregated disclosures reveal that total inventories are not evenly distributed across the individual inventory accounts. In particular, finished goods, work-in-process, and raw materials, on average, account for 35.8%, 25.6%, and 38.6% of total inventories, respectively. In turn, finished goods, work-in-process, and raw materials, on average, account for 7.7%, 5.5%, and 8.3% of total assets, respectively.

Figure 1c and d present the time series of inventory holdings. We find a decreasing trend in total inventory holdings, which suggests increasing inventory efficiencies for the average manufacturer. Specifically, the average value of *TOTHLD* decreases from 33% in 1977 to 15% in 2006. This decreasing trend is also evident for raw materials, work-in-process, and finished goods. Focusing on inventory holding periods, the average manufacturer holds finished goods for 39 days, work-in-process for 31 days, and raw materials for 44 days, while the total inventory cycle lasts for approximately 114 days. The mean value of the indicator variable *DREVERSAL* shows that 8.9% of the full sample is associated with inventory reversals. In turn, the mean value of the indicator variable *DWD* shows that 5.8% of the post-1995 subsample is associated with inventory write-downs.

Table 1, Panel B, reports pairwise correlations and yields preliminary evidence that *CC* is negatively related to total inventory holdings and inventory

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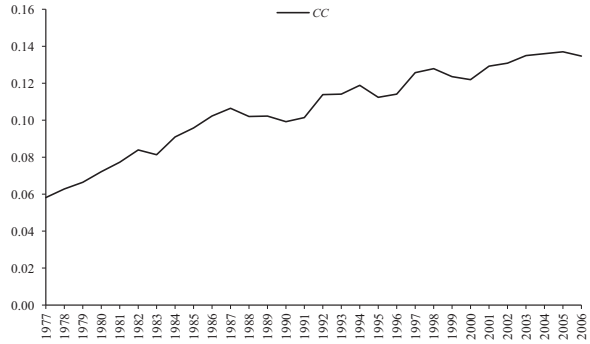
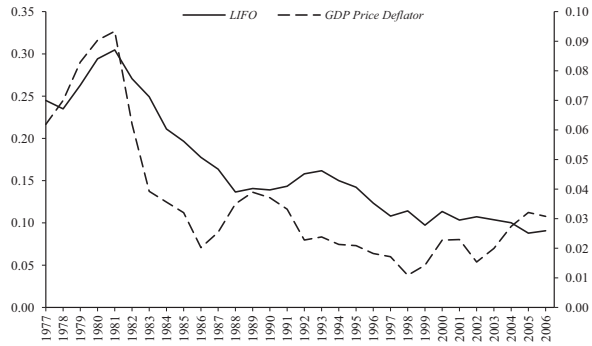
Figure 1 Time Series of Customer-Base Concentration and Inventory Variables (a) Customer-Base Concentration (b) Reported LIFO Reserve Frequency (Left Axis) and GDP Price Deflator (Right Axis) (c) Total Inventory Holdings (d) Total Inventory Holding Components

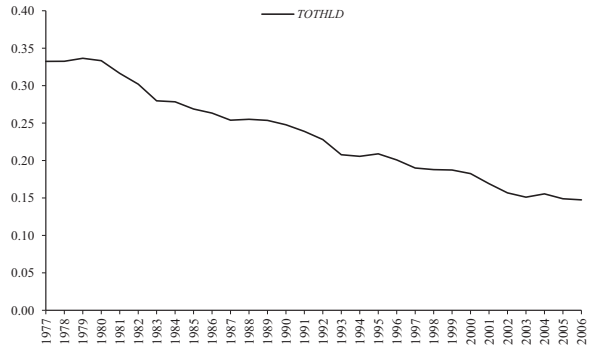
Panel (a): Customer-base concentration.



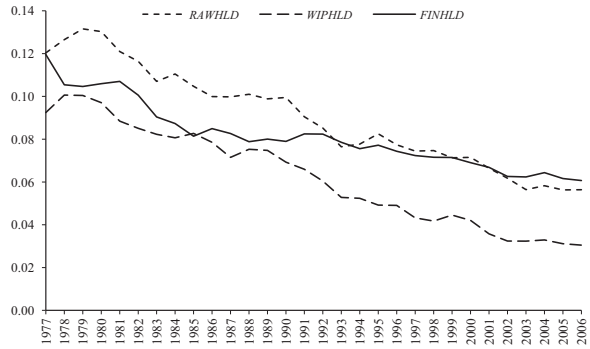
Panel (b): Reported LIFO reserve frequency (left axis) and GDP price deflator (right axis).



Panel (c): Total inventory holdings.



Panel (d): Total inventory holding components.



Notes: (a, b, c, and d) This Figure present the time series of annual cross-sectional averages for customer-base concentration (CC), reported LIFO reserve frequency (LIFO), total inventory holdings (TOTHLD), and inventory holding component, including raw materials holdings (RAWHLD), work-in-process holdings (WIPHLD), and finished goods holdings (FINHLD). The sample includes 15,084 firm-year observations from 1977 to 2006.

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holding periods as well as to inventory write-downs and reversals. Table 1, Panel C, compares the mean and median values of inventory variables for the top and bottom quintiles of *CC*. Consistent with the pairwise correlations, total inventory holdings and inventory holding periods, as well as the likelihood of inventory write-downs and reversals, are on average lower for more concentrated manufacturers.

4.2. Customer-Base Concentration and Inventory Efficiencies

Table 2 reports results from regressions of inventory holdings on *CC* based on equation (2). We find a negative association between total inventory holdings and *CC* after controlling for a wide array of firm characteristics, including firm size, age, growth, sales volatility, number of reported business segments, total leverage, gross margins, and capital intensity, as well as industry and year fixed effects. In terms of economic magnitude, all else equal, a one-standard deviation (SD) increase in *CC* is associated with a 7.03% decrease in total inventory holdings (significant at the 1% level). We next probe the link between *CC* and inventory holding components, including raw materials, work-in-process, and finished goods.

Looking across columns 2 through 4, the coefficient estimates on *CC* are −0.075, 0.021, and −0.036, for finished goods, work-in-process, and raw materials, respectively, and together add up to −0.090, which by virtue of OLS regression properties is equal to the coefficient on *CC* for total inventory holdings. In terms of the impact on total inventory holdings, all else equal, a 1 SD increase in *CC* is associated with a 5.86% decrease due to finished goods, a 1.64% increase due to work-in-process, and a 2.81% decrease due to raw material holdings. We conclude that efficiencies in terms of lower total inventory holdings for more concentrated manufacturers primarily flow through the finished goods account, while lower raw material inventories are offset by higher work-in-process holdings.

Table 3 reports results from regressions of inventory holding periods on *CC* based on equation (2). The regression results for inventory holding periods mirror those for inventory holdings. First, we find a negative association between total inventory holding period and *CC*. In terms of economic magnitude, all else equal, a 1 SD increase in *CC* is associated with a 6.26% decrease in total inventory holding period (significant at the 1% level). Second, a comparison of the coefficient estimates for the disaggregated inventory accounts reveals that evidence of shorter inventory holding cycles for more concentrated manufacturers is mostly driven by the negative association between *CC* and finished goods holding days, while shorter raw material holding periods are offset by longer work-in-process cycles. In terms of the impact on total inventory holding period, all else equal, a 1 SD increase in *CC* is associated with a 5.69% decrease due to finished goods, a 2.08% increase due to work-in-process, and a 2.65% decrease due to raw material holding days.

The coefficient estimates on the control variables are in line with prior research on the determinants of inventory turns. Inventory holding periods increases in gross margins and decrease in sales growth and capital intensity, which is consistent with prior evidence from the retail sector (e.g., Gaur and Kesavan 2009, Gaur et al. 2005). We also find that inventory

Table 2 Inventory Holding Components and Customer-Base Concentration

	Regressions of inventory holdings			
	(1) Total inventories	(2) Finished goods	(3) Work-in-process	(4) Raw materials
Intercept	0.226***	0.132***	−0.020**	0.114***
	(11.37)	(8.67)	(−2.17)	(12.93)
CC_{it}	−0.090***	−0.075***	0.021**	−0.036***
	(−6.77)	(−9.80)	(2.05)	(−5.72)
MV_{it}	−0.021***	−0.005***	−0.003***	−0.013***
	(−17.06)	(−5.85)	(−4.96)	(−18.85)
AGE_{it}	0.032***	0.011***	0.015***	0.007***
	(9.52)	(4.34)	(7.06)	(3.89)
SG_{it}	−0.007***	−0.004***	−0.002	−0.002
	(−3.41)	(−3.87)	(−1.64)	(−1.42)
SGV_{it}	−0.016*	−0.010**	−0.002	−0.004
	(−1.86)	(−2.12)	(−0.63)	(−0.99)
$CONGLO_{it}$	−0.005	−0.004	0.002	−0.003
	(−1.14)	(−1.31)	(0.87)	(−1.36)
LEV_{it}	0.001	0.000**	0.000	−0.000
	(1.19)	(2.02)	(0.34)	(−0.19)
GM_{it}	0.008***	0.003***	0.001**	0.003***
	(6.05)	(5.39)	(2.53)	(4.80)
$ACCLOSS_{it}$	0.005	0.010**	−0.003	−0.002
	(1.17)	(2.36)	(−0.94)	(−0.82)
CAP_{it}	−0.116***	−0.066***	−0.004	−0.046***
	(−5.27)	(−6.43)	(−0.46)	(−4.79)
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Adjust R^2 (%)	37	27	22	24

This Table reports results from pooled cross-sectional OLS regressions of inventory holdings, including raw materials ($RAWHLD_{it}$), work-in-process ($WIPHLD_{it}$), finished goods ($FINHLD_{it}$), as well as total inventory holdings ($TOTHLD_{it}$) on customer-base concentration (CC_{it}) and a vector of control variables, including log market value of equity (MV_{it}), log firm age (AGE_{it}), sales growth (SG_{it}), sales growth volatility (SGV_{it}), an indicator variable for multi-segment firms ($CONGLO_{it}$), leverage (LEV_{it}), gross margins (GM_{it}), accumulated losses ($ACCLOSS_{it}$), capital intensity (CAP_{it}), industry fixed effects based on two-digit SIC codes, and year fixed effects. Statistical significance is based on *t*-statistics (in parentheses) using standard errors clustered by firm and year. ***, **, and * indicate statistical significance at 1, 5, and 10% levels, respectively, using two-tailed tests. The sample includes 15,084 firm-year observations from 1977 to 2006.

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Table 3 Inventory Holding Periods and Customer-Base Concentration

	Regressions of inventory holding *perio*			
	(1) Total inventories	(2) Finished goods	(3) Work-in-process	(4) Raw materials
Intercept	102.690***	62.278***	−14.983***	55.395***
	(11.69)	(10.99)	(−2.90)	(12.18)
CC_{it}	−42.262***	−38.434***	14.032*	−17.860***
	(−4.65)	(−11.03)	(1.93)	(−5.69)
MV_{it}	−0.384	0.789**	1.438***	−2.611***
	(−0.46)	(2.03)	(2.63)	(−7.55)
AGE_{it}	2.167	0.382	3.542***	−1.757*
	(1.18)	(0.37)	(2.92)	(−1.73)
SG_{it}	−6.139***	−1.390	−2.835***	−1.914**
	(−3.55)	(−1.25)	(−4.48)	(−2.10)
SGV_{it}	3.144	1.898	1.386	−0.141
	(0.44)	(0.77)	(0.56)	(−0.04)
$CONGLO_{it}$	−12.028***	−5.180***	−2.415	−4.433***
	(−4.39)	(−3.67)	(−1.35)	(−3.26)
LEV_{it}	−0.476**	0.150	−0.251**	−0.376***
	(−2.28)	(1.44)	(−2.32)	(−3.38)
GM_{it}	16.896***	6.284***	4.537***	6.076***
	(7.63)	(7.84)	(5.71)	(7.18)
$ACCLOSS_{it}$	1.830	3.404*	−0.662	−0.912
	(0.70)	(1.84)	(−0.37)	(−0.66)
CAP_{it}	−71.459***	−35.050***	−4.716	−31.693***
	(−8.09)	(−8.55)	(−0.96)	(−6.76)
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Adjust R^2 (%)	17	12	11	18

This Table reports results from pooled cross-sectional OLS regressions of inventory days outstanding, including raw materials ($RAWDAY_{it}$), work-in-process ($WIPDAY_{it}$), finished goods ($FINDAY_{it}$), as well as total inventory days outstanding ($TOTDAY_{it}$) on customer-base concentration (CC_{it}) and a vector of control variables, including log market value of equity (MV_{it}), log firm age (AGE_{it}), sales growth (SG_{it}), sales growth volatility (SGV_{it}), an indicator variable for multi-segment firms ($CONGLO_{it}$), leverage (LEV_{it}), gross margins (GM_{it}), accumulated losses ($ACCLOSS_{it}$), capital intensity (CAP_{it}), industry fixed effects based on two-digit SIC codes, and year fixed effects. Statistical significance is based on *t*-statistics (in parentheses) using standard errors clustered by firm and year. ***, **, and * indicate statistical significance at 1, 5, and 10% levels, respectively, using two-tailed tests. The sample includes 15,084 firm-year observations from 1977 to 2006.

holding periods are shorter for firms that report at least two business segments, which is consistent with prior evidence of higher productive efficiency for multi-segment firms relative to single-segment firms (e.g., Schoar 2002).

To summarize, we find that more concentrated manufacturers tie up a smaller fraction of their assets in inventories and experience shorter inventory holding periods.[4] Our investigation of individual inventory accounts draws a nuanced picture of the link between *CC* and inventory efficiencies. Inventory efficiencies primarily flow through the finished goods inventory account, which is consistent with the focus of recent research on finished goods when exploring variation in inventory efficiencies (e.g., Cachon and Olivares 2010), while raw material efficiencies are offset by higher work-in-process holdings and longer work-in-process cycles. In general, the work-in-process inventory is determined by queuing, processing, waiting for batch, moving, and waiting to match (e.g., Hopp and Spearman 2011). With these determinants in mind, a potential explanation for longer work-in-process cycles is that processing time increases when servicing a limited number of major customers due to higher product customization to meet customers' demands. Without access to proprietary plant-level data, however, it is impossible to directly test the empirical link of *CC* with processing time and other work-in-process determinants.

4.3. Customer-Base Concentration and Inventory Write-Downs

Next, we search for a link between inventory investment efficiencies and *CC*. Our analysis is based on hand-collected data from 10-K filings about the occurrence and magnitude of inventory write-downs post-1995, that is, after the introduction of the SEC EDGAR database.

Table 4 reports logistic and Tobit regression results for inventory write-downs based on equations (4) and (5), respectively. Starting with the logistic regression results, we find that more concentrated manufacturers are significantly less likely to report inventory write-downs. All else equal, a 1 SD increase in *CC* is associated with a 44% decrease in the odds of an inventory write-down. The Tobit regression results mirror the logistic regression results. The coefficient estimate on *CC* in Column 2 is significantly negative, which suggests that more concentrated manufacturers are not only associated with a lower likelihood of inventory write-downs but also conditioning on the occurrence of an inventory write-down its absolute magnitude decreases with *CC*.

We note that inferences based on the coefficient estimates on the control variables are consistent with economic intuition. Large and high growth firms are less likely to report an inventory write-down, while manufacturers with more volatile sales and accumulated losses are more likely to report such write-downs. The positive coefficient estimate on *GM* implies a link between gross margins and the propensity of firms to end up with excess inventory, which is consistent with our evidence that manufacturers with higher gross margins hold more inventories and experience longer inventory holding periods.

To examine the association between *CC* and inventory investment efficiencies for our full sample period, that is, including the years prior to the introduction of the SEC EDGAR database, we also

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Table 4 Inventory Write-Downs and Customer-Base Concentration

	Regressions of inventory write-downs	
	(1) Logistic model	(2) Tobit model
Intercept	−2.660***	−0.102***
	(−3.66)	(−4.65)
CC_{it}	−1.485***	−0.049***
	(−2.74)	(−16.56)
MV_{it}	−0.081**	−0.004***
	(−2.21)	(−7.70)
AGE_{it}	−0.089	−0.006***
	(−1.06)	(−3.98)
SG_{it}	−3.009***	−0.110***
	(−13.61)	(−22.69)
SGV_{it}	0.542**	0.019***
	(2.27)	(9.60)
$CONGLO_{it}$	0.197	0.004*
	(1.49)	(1.71)
LEV_{it}	0.026	0.001***
	(1.38)	(6.58)
GM_{it}	0.157***	0.006***
	(2.95)	(8.48)
$ACCLOSS_{it}$	0.433***	0.015***
	(3.45)	(29.84)
CAP_{it}	−0.208	−0.019***
	(−0.39)	(−2.84)
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Pseudo R^2 (%)	16	40

This Table reports results from pooled cross-sectional logistic and Tobit regressions of the inventory write-down indicator variable (DWD_{it}) and the inventory write-down magnitude (WD_{it}), respectively, on customer-base concentration (CC_{it}) and a vector of control variables, including log market value of equity (MV_{it}), log firm age (AGE_{it}), sales growth (SG_{it}), sales growth volatility (SGV_{it}), an indicator variable for multi-segment firms ($CONGLO_{it}$), leverage (LEV_{it}), gross margins (GM_{it}), accumulated losses ($ACCLOSS_{it}$), capital intensity (CAP_{it}), industry fixed effects based on two-digit SIC codes, and year fixed effects. Goodness-of-fit is measured using McFadden's (1974) pseudo R2, which compares the unrestricted (i.e., full model) and restricted (i.e., constant only) log likelihoods. Statistical significance is based on z-statistics (in parentheses) using standard errors clustered by firm and year. ***, **, and * indicate statistical significance at 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample includes 7893 firm-year observations from 1995 to 2006.

Table 5 Inventory Reversals and Customer-Base Concentration

	Regressions of inventory reversals	
	(1) Logistic model	(2) Tobit model
Intercept	−1.754***	−0.170***
	(−4.92)	(−5.68)
CC_{it}	−0.731**	−0.059**
	(−2.50)	(−2.22)
MV_{it}	−0.098***	−0.012***
	(−3.69)	(−17.28)
AGE_{it}	−0.046	−0.008
	(−0.88)	(−1.44)
SG_{it}	−2.474***	−0.172***
	(−12.37)	(−7.15)
SGV_{it}	0.379*	0.052**
	(1.80)	(2.11)
$CONGLO_{it}$	0.067	0.001
	(0.86)	(0.08)
LEV_{it}	0.014	0.001***
	(1.19)	(14.35)
GM_{it}	0.143***	0.011***
	(3.09)	(2.68)
$ACCLOSS_{it}$	0.354***	0.031***
	(4.76)	(4.14)
CAP_{it}	0.165	0.000
	(0.80)	(0.03)
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Pseudo R^2 (%)	13	20

This Table reports results from pooled cross-sectional logistic and Tobit regressions of the inventory reversal indicator variable ($D_REVERSAL_{it}$) and the inventory reversal magnitude ($REVERSAL_{it}$), respectively, on customer-base concentration (CC_{it}) and a vector of control variables, including log market value of equity (MV_{it}), log firm age (AGE_{it}), sales growth (SG_{it}), sales growth volatility (SGV_{it}), an indicator variable for multi-segment firms ($CONGLO_{it}$), leverage (LEV_{it}), gross margins (GM_{it}), accumulated losses ($ACCLOSS_{it}$), capital intensity (CAP_{it}), industry fixed effects based on two-digit SIC codes, and year fixed effects. Goodness-of-fit is measured using McFadden's (1974) pseudo R^2, which compares the unrestricted (i.e., full model) and restricted (i.e., constant only) log likelihoods. Statistical significance is based on z-statistics (in parentheses) using standard errors clustered by firm and year. ***, **, and * indicate statistical significance at 1, 5, and 10% levels, respectively, using two-tailed tests. The sample includes 15,084 firm-year observations from 1977 to 2006.

search for a link between inventory reversals and *CC*. As explained in subsection 3.3, we define an inventory reversal as a decline in total inventories in excess of 1% of the beginning of period total assets accompanied by commensurate negative Compustat special items.

Table 5 reports logistic and Tobit regression results for inventory reversals based on equations (4) and (5), respectively. Focusing on the logistic regression results, the coefficient estimate on *CC* is significantly negative while its magnitude implies that, all else equal, a 1 SD increase in *CC* is associated with a 47% decrease in the odds of an inventory reversal. Turning to the Tobit regression results, the coefficient estimate on *CC* is also significantly negative, which suggests that more concentrated manufacturers are not only less likely to experience an inventory reversal but also conditioning on recording an inventory reversal its absolute magnitude decreases with *CC*.[5]

To summarize, our logistic and Tobit regression analysis of inventory write-downs and reversals yields evidence consistent with a positive link between *CC* and inventory investment efficiencies. Manufacturers with more concentrated customer bases are associated with a lower likelihood and magnitude of inventory write-downs and reversals, which is consistent with the OM view that manufacturers can achieve inventory efficiencies through relationships with a limited number of major customers.

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4.4. Customer-Base Concentration and Firm Valuation

The findings presented so far provide evidence of inventory efficiencies for more concentrated manufacturers. Next, we test whether more concentrated manufacturers are associated with a valuation premium. Table 6 reports results from regressions of Tobin's Q on CC based on equation (6). Consistent with the pairwise correlations, we find a positive association between Tobin's Q and CC after controlling for other firm characteristics as well as industry and year fixed effects. All else equal, a 1 SD increase in CC implies a 13.30% increase in Tobin's Q (significant at the 1% level). Column 2 extends the right-hand side of equation (6) to include the inverse of total assets ($1/TA$), that is, the scale variable of the Tobin's Q, and shows that evidence of a positive link between CC and firm valuation is not sensitive.

Our evidence of a valuation premium for more concentrated manufacturers is related to recent research on the implications of major customer dependency for cost of capital, default risk, and firm valuation (e.g., Campello and Gao 2014, Cen et al. 2015, Dhaliwal et al. 2015, Osadchiy et al. 2015).[6] Columns 3 and 4, however, show that our evidence is not sensitive after extending the right-hand side of equation (6) to include an implied cost of capital estimate (COC) and Merton's (1974) distance-to-default (DD), which is a market-based measure of default risk.[7]

Turning to the coefficient estimates on the control variables, the evidence is in line with prior research on firm valuation. Large and high growth firms are rewarded with higher valuation multiples, while older and more levered firms are associated with lower multiples. The negative slope on CAP suggests a negative premium for higher capital intensity, while the negative slope on $CONGLO$ is consistent with long-standing evidence of a corporate diversification discount (e.g., Lang and Stulz 1994). Finally, the significantly negative coefficient on COC is consistent with the valuation property that higher cost of capital translates into lower firm valuation. We hasten to note that evidence of a valuation premium for more concentrated manufacturers may only be partially attributed to the positive link between CC and inventory efficiencies. Nevertheless, the evidence suggests that investors trade off the costs and benefits of major customer relationships and, on balance, consider CC as a net positive for firm valuation.

Table 6 Firm Valuation and Customer-Base Concentration

	Regressions of Tobin's Q			
	(1)	(2)	(3)	(4)
Intercept	1.067***	−0.178	−0.003	0.022
	(5.10)	(−0.75)	(−0.01)	(0.09)
CC_{it}	1.425***	1.316***	1.315***	1.314***
	(5.13)	(4.92)	(4.91)	(4.89)
MV_{it}	0.275***	0.457***	0.457***	0.457***
	(8.09)	(10.82)	(10.81)	(10.84)
AGE_{it}	−0.342***	−0.303***	−0.303***	−0.302***
	(−8.34)	(−7.86)	(−7.86)	(−7.84)
SG_{it}	0.522***	0.481***	0.480***	0.479***
	(8.90)	(9.39)	(9.35)	(9.38)
SGV_{it}	−0.047	0.008	0.003	0.006
	(−0.42)	(0.07)	(0.02)	(0.06)
$CONGLO_{it}$	−0.222***	−0.221***	−0.221***	−0.223***
	(−3.55)	(−3.71)	(−3.71)	(−3.75)
LEV_{it}	−0.030***	−0.027***	−0.027***	−0.027***
	(−5.01)	(−4.85)	(−4.82)	(−4.75)
GM_{it}	0.003	−0.015	−0.016	−0.016
	(0.08)	(−0.50)	(−0.51)	(−0.52)
$ACCLOSS_{it}$	−0.254***	−0.060	−0.056	−0.056
	(−3.60)	(−0.91)	(−0.84)	(−0.85)
CAP_{it}	−0.854***	−0.542***	−0.544***	−0.550***
	(−4.53)	(−3.07)	(−3.08)	(−3.11)
$1/TA_{it}$		7.957***	7.971***	7.976***
		(8.85)	(8.86)	(8.86)
COC_{it}			−2.016**	−2.028**
			(−2.48)	(−2.45)
DD_{it}				−0.183
				(−0.55)
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Adjust R^2 (%)	24	31	31	31

This Table reports results from pooled cross-sectional OLS regressions of Tobin's Q ($TOBIN_{it}$) on customer-base concentration (CC_{it}) and a vector of control variables, including log market value of equity (MV_{it}), log firm age (AGE_{it}), sales growth (SG_{it}), sales growth volatility (SGV_{it}), an indicator variable for multi-segment firms ($CONGLO_{it}$), leverage (LEV_{it}), gross margins (GM_{it}), accumulated losses ($ACCLOSS_{it}$), capital intensity (CAP_{it}), the inverse of total assets ($1/TA_{it}$), Easton's (2004) implied cost of capital (COC_{it}), Merton's (1974) distance-to-default (DD_{it}), industry fixed effects based on two-digit SIC codes, and year fixed effects. Statistical significance is based on t-statistics (in parentheses) using standard errors clustered by firm and year. ***, **, and * indicate statistical significance at 1, 5, and 10% levels, respectively, using two-tailed tests. The sample includes 15,084 firm-year observations from 1977 to 2006.

5. Conclusion and Directions for Future Research

We investigate the link between CC and inventory efficiencies in the manufacturing sector. We find that more concentrated manufacturers hold fewer inventories for less time and are less likely to end up with excess inventories, as indicated by the lower likelihood and magnitude of inventory write-downs and reversals. Using disaggregated inventory disclosures, we find that inventory efficiencies primarily flow through the finished goods inventory account, while raw material efficiencies are offset by higher work-in-process holdings and longer work-in-process cycles. We also document a valuation premium for more concentrated manufacturers, which implies that investors trade off the costs and benefits of relationships with a

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limited number of major customers and, on balance, consider *CC* as a net positive for firm valuation.

Taken together, our evidence is consistent with the OM view that manufacturers can achieve inventory efficiencies through relationships with a limited number of major customers. An interesting direction for future research would be to examine the implications of single vs. multiple sourcing strategies for inventory efficiencies. Given that firms are only required to disclose information about their major customers, a challenge for future researchers will be the identification of supplier firms with single vs. multiple vendors. Another direction for future research would be to search for variation in our findings across industries within the manufacturing sector as well as to further disaggregate the drivers of inventory efficiencies using plant-level data. Such research would further broaden our knowledge of the link between customer-base structure, fundamental analysis, and firm valuation.

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Appendix A: Labels and Definitions of Variables

Label	Definition
CC_{it}	Customer-base concentration measured as in Patatoukas (2012)
$LIFO_{it}$	Indicator variable = 1 if the firm reported LIFO reserve; = 0 otherwise
$TOTHLD_{it}$	Total inventories scaled by total assets
$RAWHLD_{it}$	Raw materials inventories scaled by total assets
$WIPHLD_{it}$	Work-in-process inventories scaled by total assets
$FINHLD_{it}$	Finished goods inventories scaled by total assets
$TOTDAYS_{it}$	Total inventories scaled by CoGS multiplied by 365
$RAWDAYS_{it}$	Raw materials inventories scaled by CoGS multiplied by 365
$WIPDAYS_{it}$	Work-in-process inventories scaled by CoGS multiplied by 365
$FINDAYS_{it}$	Finished goods inventories scaled by CoGS multiplied by 365
ΔINV_{it}	Change in total inventories scaled by beginning of year total assets

(continued)

Appendix A: ***(Continued)***

Label	Definition
SI_{it}	Compustat special items scaled by the beginning of year total assets
$DREVERSAL_{it}$	Indicator variable = 1 if $\Delta INV_{it} \leq -0.01$ and $SI_{it} \leq -0.01$;= 0 otherwise
$REVERSAL_{it}$	Product of $D_REVERSAL_{it}$ multiplied by the absolute value of SI_{it}
DWD_{it}	Indicator variable = 1 for inventory write-downs; 0 = otherwise
WD_{it}	Absolute value of inventory write-downs
MV_{it}	Log of market value of equity as of fiscal year-end
$TOBIN_{it}$	Tobin's *Q* ratio measured as market value of equity plus (short-term and long-term) debt scaled by total assets
AGE_{it}	Log of firm age measured relative to first year of Compustat coverage
SG_{it}	Annual percentage growth in sales
SGV_{it}	Sales growth volatility measured as the cross-sectional standard deviation of annual percentage growth in sales across annual partitions based on two-digit SIC codes
$CONGLO_{it}$	Indicator variable = 1 if the firm reports at least two business segments
LEV_{it}	Total leverage measured as book value of total assets scaled by the book value of equity
GM_{it}	Gross margin measured as one minus CoGS scaled by sales
$ACCLOSS_{it}$	Indicator variable = 1 if at the beginning of the year the firm reported book value of equity that is either negative or below market value
CAP_{it}	Net PP&E scaled by total assets
$1/TA_{it}$	The inverse of total assets
DD_{it}	Distance-to-default measured across annual partitions based on two-digit SIC codes as the cross-sectional average value of Merton's (1974) distance-to-default measure of default risk using the SAS code of Bharath and Shumway (2008)
COC_{it}	Cost of capital measured across annual partitions based on two-digit SIC codes as the cross-sectional average value of the implied cost of capital estimate using Easton's (2004) modified PEG ratio mode.

Notes

[1]Our inferences are not sensitive when we estimate the models using annual cross-sectional regressions and base statistical inferences on Fama and MacBeth (1973) *t*-statistics. Our inferences are also not sensitive when we use random coefficients regression models allowing for heterogeneity in the coefficient estimates over time and across firms. In additional tests, we find consistent results using rank and log transformations of *CC*, whereas including second and higher order polynomial transformations of *CC* does not yield significant evidence of nonlinear relations. These results are available upon request.

[2]In general, there are two approaches for reporting inventory write-downs: the reserve and the direct approach. Under the reserve approach, the firm maintains an inventory reserve account that is increased as a write-down is recorded and is decreased subsequently when inventories for which the reserve was created are disposed. The initial

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increase in the inventory reserve represents the expected decline in the value of inventories and the corresponding dollar amount is typically disclosed in Financial Statements Schedule II—Valuation and Qualifying Accounts at the end of the 10-K filing. Under the direct approach, the firm does not maintain a regular reserve account but rather it records a write-down expense, which is typically included in cost of goods sold and the corresponding dollar amount is disclosed in the notes to the financial statements. We note that under either approach firms typically do not associate write-downs with individual inventory accounts.

[3]An implicit assumption in the Tobin's Q calculation is that due to mark-to-market accounting the book value of debt is close to the market value of debt. Another implicit assumption is that the book value of total assets is a good proxy for the replacement cost of assets. Although both assumptions may be violated, we have no reason to believe that measurement error in Tobin's Q is correlated with *CC*.

[4]In additional tests, we exploit within-firm variation and find consistent evidence that customer-base concentration increases are associated with lower inventory holdings and shorter inventory holding periods. We also search for variation across partitions based on (i) supplier firm sales volatility and (ii) customer-base sales volatility and we do not find evidence of significant differences in our results. These results are available upon request.

[5]In additional tests, we search for an intertemporal link between *CC* and inventory investment efficiencies. Using logistic and Tobit regressions, we find consistent evidence that *CC* negatively predicts inventory write-downs and reversals across forecast horizons ranging from one to three years ahead. These results are available upon request.

[6]Cen et al. (2015) argue that major customers have strong incentives to evaluate and maintain the stability of their supply chain, and thus the existence and continuation of long-term major customer relationships provide a *de facto* certification to the suppliers as firms of higher operating stability and lower default risk. Campello and Gao (2014) argue that in spite of the beneficial effects of major customer dependency on supplier performance, higher levels of concentration increase interest rate spreads and the number of restrictive covenants on a firm's bank loans, while reducing loan maturities. Dhaliwal et al. (2015) provide evidence that implied cost of capital estimates as well as measures of systematic risk and idiosyncratic risk increase with major customer dependency. Osadchiy et al. (2015) find evidence of an inverse relation between major customer dependency and measures of systematic risk, which they attribute to risk-reduction benefits from order pooling across major customers.

[7]We estimate distance-to-default using Bharath and Shumway's (2008) SAS code. We estimate implied cost of capital using Easton's (2004) modified PEG ratio model (our results are not sensitive using Dhaliwal et al.'s (2015) composite index of alternative cost of capital estimates). Due to the additional data requirements, we measure distance-to-default and implied cost of capital using annual cross-sectional averages across two-digit SIC codes.

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